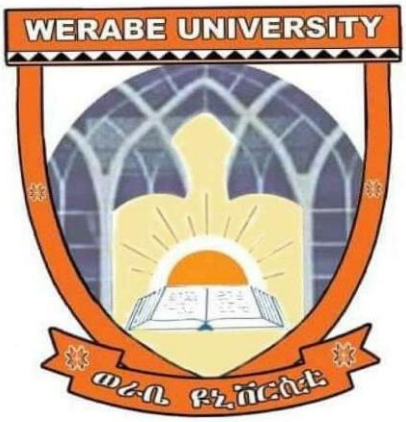


WERABEUNIVERSITY

COLLEGE OF INSTITUTIONS OF TECHNOLOGY

DEPARTMENT OF *COMPUTER SCIENCE*

Softwareengineering course

Projecttitle:WerabeuniversityOnlineExaminationSystem(WROES)

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SoftwareDevelopmentPlan(SDP)

ProjectTitle: ***Werabeuniversity*** OnlineExaminationSystem

1. Introduction

The introduction sets the stage for the project by outlning its purpose and significance.The ***Werabeuniversity*** OnlineExaminationSystemaims to enhance the examination process by providing a secure, user-friendly platform for students and faculty. Key points to include:

- ***Background:***Discussthetraditionalexaminationmethodsandtheirlimitations,suchas logistical challenges, time constraints, and the potential for academic dishonesty.
- ***Objectives:***Defineclearobjectivessuchasimprovingaccessibility,increasingefficiency in exam administration, and providing immediate feedback to students.
- ***Benefits***:Highlightthebenefitsforstakeholders,includingeaseofuse,reducedpaperusage, and the ability to analyze exam results quickly.

2. ProjectOrganization

Thissectiondetailsthestructure oftheprojectteam:

RolesandResponsibilities:Clearlydefinerolessuchas:

- ✓ ***ProjectManager:***Overseestheproject,ensuringitstaysontrackandwithinbudget.
- ✓ ***Developers:***Responsibleforcodingandimplementingsystemfeatures.
- ✓ ***UI/UXDesigners:***Focusoncreatinganintuitiveuserinterfaceandenhancinguser experience.
- ✓ ***Testers:***Conductrigoroustestingtoensurethesystemisfunctional,secure,anduser-friendly.
- ✓ ***Stakeholders:***Includefacultymembers,ITstaff,andstudentrepresentativeswhowill provide input throughout the development process.

CommunicationPlan:Establishregularcommunicationchannels(meetings,reports,tools)to ensure all team members and stakeholders are aligned.

3. ProjectSchedule

Theprojectscheduleoutlinesthetimelinefordevelopment:

PhasesofDevelopment:

- ❖ ***Planning:***Initialmeetings,requirementgathering,andfeasibilitystudies.
- ❖ ***Design:***Creatingwireframesandprototypes,definingtechnicalspecifications.
- ❖ ***Implementation:***Actualcoding,databasesetup,andintegrationoffeatures.
- ❖ ***Testing:***Systematictestingphasestoidentifyandfixbugs.
- ❖ ***Deployment:***Launchingthesystemforliveuse.

- ❖ ***Maintenance:*** Ongoing support and updates after deployment.

- **Milestones**: Define key milestones and deadlines for each phase to track progress.

4. Resources

This section details necessary resources:

- **Human Resources**: Identify the required skillsets and team members for each role.
- **Technical Resources**: Outline hardware (servers, devices for testing) and software (development environments, testing tools) needed for the project.
- **Financial Resources**: Provide an estimated budget covering personnel, tools, software licenses, and unforeseen costs, ensuring adequate funding is available for all phases of development.

5. Risk Management

Effective risk management is critical for project success:

- **Risk Identification**: List potential risks such as:
  - ✓ ***Technical Risks:*** Challenges in integrating new technologies or unforeseen coding issues.
  - ✓ ***Operational Risks:*** Resource availability, including team members leaving the project.
  - ✓ ***Schedule Risks:*** Delays in development or testing phases.
  - ✓ ***Financial Risks:*** Budget overruns or funding shortfalls.
- **Risk Analysis**: Evaluate the likelihood and impact of each risk, categorizing them as high, medium, or low.
- **Mitigation Strategies**: Develop plans to address each risk, including contingency plans for high-impact risks. Regularly review and update this section throughout the project.
- Implement redundant servers and backup systems.
- Use secure authentication and encryption to prevent breaches.

6. Development Methodology

Choosing the right development methodology is crucial:

- **Agile Methodology**: Consider using Agile for its flexibility and iterative approach, allowing for frequent reassessment and adaptation of plans based on feedback.
  - ✓ ***Sprints:*** Organize work into 2–4-week sprints, focusing on delivering small, functional increments of the system.
  - ✓ ***Stakeholder Engagement:*** Involve stakeholders in sprint reviews to gather feedback and adjust requirements as necessary.
- **Process:** Define steps like requirements analysis, system design, coding, testing, and deployment, tailored to the chosen methodology.
- **Documentation**: Emphasize the importance of maintaining documentation throughout the

development process to facilitate knowledge sharing.

7. QualityAssurance

Qualityassuranceensuresthesystemmeetsthedesiredstandards:

- **TestingStrategies**:
  - ***UnitTesting***:Testindividualcomponentsforcorrectness.
  - ***IntegrationTesting***:Ensurethatcomponentsworktogetherasexpected.
  - ***SystemTesting***:Validatethecompletesystemagainstrequirements.
  - ***UserAcceptanceTesting(UAT)***:Involveactualusersintestingtoconfirm thatthe system meets their needs.
- **ContinuousImprovement**:Implementafeedbackloopwherefindingsfromtestinginform future development and refinement processes.

8. ConfigurationManagement

Configurationmanagementmaintainscontroloversystemchanges:

- **VersionControl**:UtilizesystemslikeGitfortrackingchangesincodeanddocumentation, enabling collaboration among team members.
- **ChangeManagementProcess**:Establishaformalprocedureforproposing,reviewing,and implementing changes to the project scope or requirements.
- **Documentation**:Maintainup-to-datedocumentationonconfigurationitems,including software versions, system architecture, and deployment processes.
- **Tools**:UseGitHub,GitLab,orBitbucketforcoderepositoriesandtoolslikeAnsiblefor server configuration.
- **BackupandRecovery**:Maintainbackupsofcodeanddatatorecoverfromfailures.

9. Documents

Documentationisacriticalcomponentofanysoftwareproject:

- **TypesofDocumentation**:
  - ***RequirementsDocument***:Definessystemrequirementsandspecifications.
  - ***DesignDocuments***:Detailedarchitectureanddesigndecisions.
  - ***UserManuals***:Guidesforstudentsandfacultyonhowtousethesystem.
  - ***TechnicalDocumentation***:Forfuturedeveloperstounderstandthecodebaseand system architecture.
- **DocumentationManagement**:Establishacentralizedrepositoryforallprojectdocumentation, ensuring it is accessible and organized for easy reference.
- **Purpose**:Ensurestakeholdershaveclear,up-to-dateinformationandthesystemcanbe maintained post-deployment.

- **Storage**:Storedocumentsinacentralizedrepository(e.g.,GoogleDrive,Confluence)with version control.

10. ProjectMonitoringandControl

Monitoringandcontrolhelpensuretheprojectstaysontrack:

- **PerformanceMetrics**:Establishkeyperformanceindicators(KPIs)suchas:
  - ***ScheduleVariance:***Measureiftheprojectisonschedule.
  - ***BudgetVariance:***Trackifspendingalignswiththebudget.
  - ***QualityMetrics:***Assessdefectratesandusersatisfaction.
- **RegularStatusReports**:Prepareweeklyorbi-weeklyreportstocommunicateprogressto stakeholders, highlighting achievements, challenges, and next steps.
- **Adjustments**:Bepreparedtomakeadjustmentsasneededbasedonmonitoring results,ensuring the project remains aligned with its objectives.
- **Metrics:**Trackkeyperformanceindicators(KPIs)likepercentageoftaskscompleted,defect rates, and stakeholder satisfaction.

Chapter One:

SoftwareDevelopment

Plan

1.1 Background

The Werabe university Online Examination System (WROES) is designed to modernize the traditional paper-based examination process by making it digital and automated. Right now, the university struggles to manage exams effectively due to manual tasks such as exam creation, distribution, grading, and reporting results. The proposed system aims to simplify these tasks by offering a web-based platform. This platform will allow instructors to create, manage, and grade exams online. It will also enable students to register, take exams, and view results digitally.This project supports Werabe University's goal of improving academic services through modern information technology.

1.2 ReviewofRelatedWork(LiteratureReview)

Manyuniversitieshaveswitchedtoonlineexaminationsystemstoreplacemanualtestingmethods. For example:

Moodleoffersonlinequizzesandassessmentsbutneedstechnicalsetupandlacksexam-specific analytics.

GoogleFormsprovidessimplequizcreationbutdoesnothavesecureauthenticationor automated grading for complex questions.

Other custom-built systems show progress, but they often do not meet security and scalabilityneeds.

Therefore,WROESiscreatedtoblenduser-friendliness,security,andflexibility,specifically tailoredforWerabeUniversity'sacademicstructure.

1.3 StatementoftheProblem

Werabe University currently relies on a manual exam process. This leads to time-consuming exam preparation and grading, inaccurate or delayed results, a heavy administrative workload, and an absence of centralized digital records. To solve these issues, a web-based Online Examination System is required to provide efficient and secure automated exam management, grading, and reporting features.

1.4 Objectives

1.4.1 GeneralObjective

To design and develop a web-based Online Examination System for Werabe University that automates the exam process from creation to grading and reporting results.

1.4.2 SpecificObjectives

- ✓ Toallowstudents,instructors,andadministratorstoregisterandlogintothesystem.
- ✓ Toenableinstructorstocreate,schedule,andmanageexams.
- ✓ Tofacilitateautomaticgradingforobjective-typequestions.
- ✓ Togivestudentsinstantresultsandperformancereports.
- ✓ Toassistadministratorswithreportgenerationandsystemmanagement.
- ✓ Toensuredatasecurity,efficiency,andreliability.

1.5 Methodology

1.5.1 DataCollectionMethodology

Informationforthissystemwasgatheredthrough:

- ✓ Interviewswithinstructors
- ✓ ITstaff,and
- ✓ Examofficers
- ✓ Questionnaireswheredistributedtostudentsand
- ✓ Stafftoassesstheirneeds.

Wealsoobserved currentexammanagement practicesand reviewed universityexampoliciesand records.

1.5.2 SystemDevelopmentMethodology

We will use the Incremental Software Development Model to develop WROES. This model divides the system into increments, with each increment delivering a working version. Each increment goes through all stages of the software development life cycle: requirement gathering, design, coding, testing, and deployment. Users can access early modules while developers add new features.

The key benefits of the Incremental Model include modularity, flexibility, early feedback, reduced risk, and easier testing.

WhytheIncrementalModelIstheBestChoiceforWROES

Reason	JustificationforWROES
ModularDevelopment	WROESconsistsofindependentmodules(login,exam creation, grading, reports) developed separately.
FlexibilityinRequirements	University policies and user needs may change;updates can be made without affecting previous increments.
EarlyDelivery	Studentsandinstructorscanusefeaturessuchasregistration before the full system is completed.
ContinuousUserFeedback	Eachincrementallowsrealuserstotestandprovide feedback.
EasierTestingandMaintenance	Each module is tested independently, making debugging simple.
ReducedRisk	Errorsaredetectedearlybeforefulldeployment.
EfficientTeamwork	Differentteammemberscan developincrements simultaneously.
IdealforWebSystems	Supportsscalabilityandregularupdatesforacademic systems.

1.5.3 SystemDevelopmentTools

- Category Tool/Technology
- Frontend HTML,CSS,JavaScript,Bootstrap
- Backend PHP/Node.js

- Database MySQLorOracle
- ServerApache (XAMPP)

DesignToolsFigma,Draw.io,ERDandDFDtools

1.6 Scope(KeyFunctionsoftheSoftware)

The system includes:

- Securelogin
- Exammanagement
- Automatedgrading
- Reporting

Itdoesnotcurrentlysupportofflineexamsormobileappintegration(thesefeaturesareplanned for the future).

1.7 TeamCompositionandResponsibilityAssignmentMatrix

MemberName	Role	Responsibilities
TofikeMohammed	ProjectManager	Oversees the project, ensures progress, manages schedule and communication.
RediAdem	Developer	Implementsandintegratessystem modules.
NuredinKedir	UI/UXDesigner	Designs intuitive interfaces and ensures usability.
EkramMohammed	Tester/QAEngineer	Conductstestingtoensurereliability and security.
HawaAyalew	StakeholderRepresentative	Providesfeedbackandrepresentsend users.

Note:Projectsuccessdependsoncommunicationandcollaborationamongallteammembers.

1.8 WorkBreakdown

No	Task	Description	Duration
1	RequirementAnalysis	Gatheranddocumentrequirements	1 week
2	SystemDesign	Developdiagrams(ERD,DFD)andUI mockups	1 week

3	Increment1Development	Build user registration and authentication	2weeks
4	Increment2Development	Developexamcreationandquestion management	2weeks
5	Increment3Development	Implementonlineexamandgrading	2weeks
6	TestingandIntegration	Conductunitandsystemtesting	1 week
7	Final Deployment & Documentation	Deployandpreparedocumentation	1 week

CHAPTER2

UseCaseModel

Introduction

The Werabe University Online Examination System (WR-OES) is designed to automate the entire lifecycle of examinations within the university environment. The system enables instructors to create, schedule, and manage examinations; allows students to take exams online; and provides administrative oversight for security, authentication, and result processing.

This Use Case Model provides a comprehensive analysis of system interactions, actors, functionalities, workflows, and internal messaging behavior. It describes how the system should operate from the viewpoint of users and external components.

PurposeoftheUseCaseModel

TheUseCaseModelservesthefollowingpurposes:

1. CapturesallfunctionalrequirementsoftheWR-OES.
2. Representssystembehavioratahighabstractionlevel.
3. Identifiestheactorsandhowtheyinteractwiththesystem.
4. Formsthefoundationfordetailedanalysis,design,testing,andimplementation.
5. Ensuressharedunderstandingbetweenstakeholders—students,instructors,system administrators, and developers.

ActorsoftheOnlineExaminationSystem

Student

Auserwhologsin,viewsavailableexams,attemptsassignedexams,submitsthem,and viewsresults.

Instructor

Responsibleforcreatingexamquestions,schedulingexams,supervisingexamsessions,andreviewing results.

Administrator

Handlessystemconfiguration,managesusers,ensuressecureexamdelivery,andoverseesexam and result records.

AuthenticationService(ExternalSystemorSubsystem)

Handlesloginvalidationandidentityverification.

Result Processing Engine (Internal System Component)

Automaticallycalculatesresultsbasedonpredefinedmarkingschemas.Key

Use Cases

ThefollowingusecasesrepresentthecompletefunctionalityoftheHU-OES.

UserLoginManageUserAccounts(Admin)
Create Examination (Instructor)
Create Questions (Instructor)
ScheduleExamination(Instructor)

PublishExamination(Instructor)
EnrollStudentsforExam(Instructor/Admin)
View Available Exams (Student)
Start Examination (Student)
Attempt Questions (Student)
Submit Examination (Student)
Auto-Grade Exam (System)
PublishResults(Instructor/Admin)
View Results (Student)
GenerateReports(Admin/Instructor)
Manage Exam Security (Admin)

UseCaseDescriptions

Belowarethefullexpandeddescriptions,suitableforacademicdocumentation.

UseCase:UserLogin

Actor:Student/Instructor/AdminToaccessthesystemsecurely. Theuser provides credentials,whichare validated bythe authenticationservice.If valid,theuser is redirected to their respective dashboard.

MainFlow:

a) Useropensloginpage.
b) Userentersusernameandpassword.
c) Systemsendsdatatoauthenticationservice.
d) Servicevalidatescredentials.
e) Systemlogsinuserandloadsrole-basedinterface.

Extensions:
– Invalid credentials → error message shown.
– Accountblocked→userreferredtoadministrator.

UseCase:ManageUserAccounts

Actor: Administrator Manage system users.
Admin creates, updates, and deactivates accounts for students and instructors.

UseCase:CreateExamination

***Actor*:**Instructor
Defineexammetadata.

MainFlow:

Instructorselects“CreateNewExam.”

Providestitle,coursecode,duration,instruction,examtype.System

stores exam in draft mode.

UseCase:CreateQuestions

Actor:Instructor
Instructorcomposesquestions:MCQ,True/False,Essay.

1. ***Extensions:***
   –Importquestionbank.
   –Attachimagesorformulas.

UseCase:ScheduleExamination

Actor:Instructor
Instructorsetsdate,time,duration,andeligiblestudentgroups.

UseCase:PublishExamination***Actor:***Instructor
Movesexamfrom“Draft”to“Published.”Studentscanviewbutnotattemptuntilstarttime.

UseCase:EnrollStudentsforExam

Actor:Instructor/Admin
Systemusescourseregistrationdatabaseormanualselection.

UseCase:ViewAvailableExams

Actor:Student
Viewlistofupcomingandactiveexams.

UseCase:StartExamination Actor:Student
Systemchecksexamschedule,authentication,attemptstatus,anddevicelockpoliciesbefore access.

UseCase:AttemptQuestions

Actor:Student
Studentsnavigatebetweenquestions,selectorwriteanswers.Auto-saverunseveryfew seconds.

UseCase:SubmitExamination

Actor: Student
Examissubmittedmanuallyorautomaticallywhentimeexpires.

UseCase:Auto-GradeExam

Actor: System
MCQsandobjectivequestionsaregradedautomaticallybytheResultProcessingEngine.

UseCase:PublishResults

Actor: Instructor/Admin
Afterverification,resultsaremadevisibletostudents.

UseCase:ViewResults

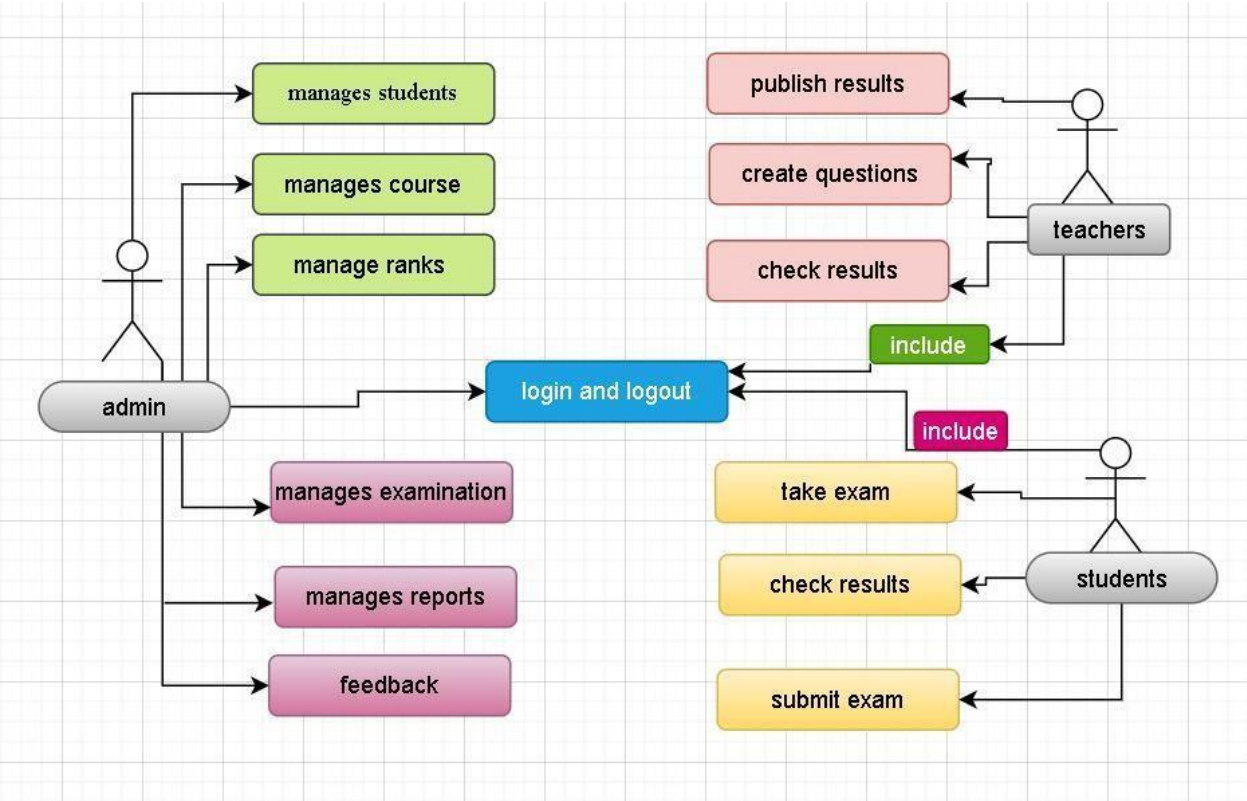
Actor: Student
Studentaccessesscorecardandfeedback(ifenabled).

UseCase:GenerateReports

2. **Actor:**Instructor/Admin
   Generatesperformancestatistics,attendancelogs,andexamactivityreports.

UseCase:ManageExamSecurity

Actor: Administrator
EnsuresIPrestrictions,preventsmultiplelogins,monitorssuspiciousactivities.



1. ClassDiagram

Purpose: Showssystemstructure:classes,attributes,methods,andrelationships.

MainClasses

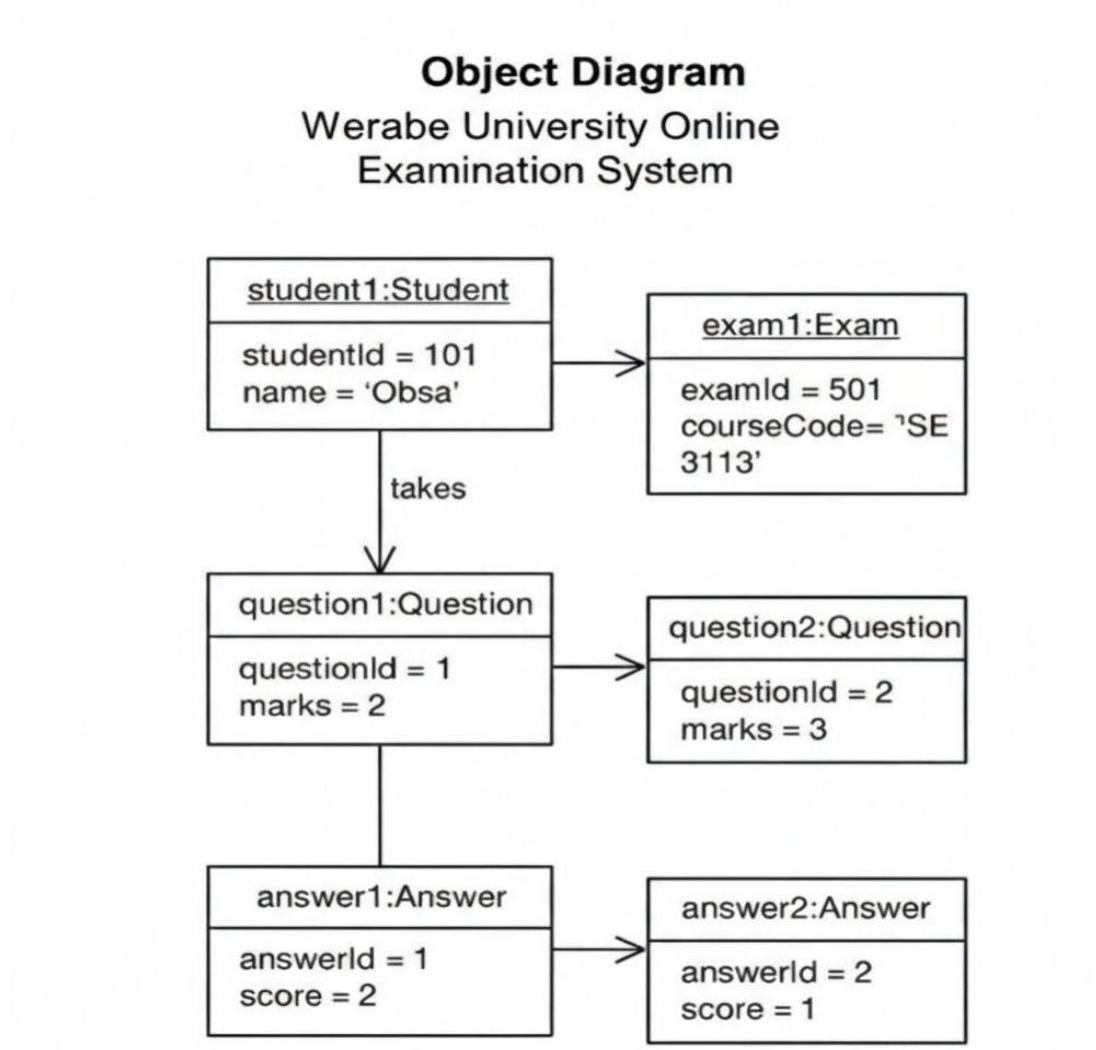
1. **Student**
   - studentId,name,email,department
   - login(),viewExam(),takeExam(),viewResult()
2. **Instructor**

- instructorId,name,department
- uploadExam(),manageQuestions(),publishResult()

3. **Admin**
    - adminId,name
    - createAccount(),manageExamSchedule()
4. **Exam**
    - examId,courseCode,examDate,duration
    - startExam(),endExam()
5. **Question**
    - questionId,questionText,questionType,marks
    - updateQuestion()
6. **Result**
    - resultId,studentId,examId,totalScore,grade
    - calculateGrade()

Relationships

- Student—takes→Exam(many-to-manyviaAnswer)
- Instructor—creates→Exam(one-to-many)
- Exam—contains→Question(composition,1..*)
- Question—isansweredwith→Answer(1..*)
- Student—has→Result(1..*)
- Admin—manages→Instructor&Studentaccounts



2. ObjectDiagram

Object Diagramssometimes referred to as Instance diagrams, are very similar toclass diagrams. Like class diagrams, they also show the relationship between objects, but they use real-world examples.

They show what a system will look like at a given time. Because there is data available in the objects, they are used to explain complex relationships between objects

Chapter3

PROPOSEDSYSTEM

3.1 Introduction

This chapter presents the **proposed system** for the Werabe University Online Examination System (WR-OES). The proposed system describes how the identified problems in the existing examination process will be addressed using a web-based solution. It outlines the overall system concept, functional and non-functional requirements, constraints, and operational scenarios.

The WR-OES is designed to automate the entire examination lifecycle, including exam creation, scheduling, online delivery, grading, result publication, and reporting. The system emphasizes security, reliability, scalability, and usability while aligning with Werabe University's academic policies and infrastructure.

3.2 OverviewoftheProposedSystem

The proposed system is a centralized, web-based Online Examination System that supports three main user roles: **Students, Instructors, and Administrators**. Each role has clearly defined responsibilities and system access privileges.

The WR-OES replaces manual and paper-based examination activities with automated digital processes. Instructors can create and manage exams, students can take exams online within defined schedules, and administrators can oversee system operation and security. The system ensures accurate record-keeping, reduces administrative workload, and provides timely feedback to students.

3.3 FunctionalRequirements

Functional requirements describe what the system must do.The primary functional requirements of WR-OES are outlined below.

Primaryfunctionalrequirementstheproposedsystemmustsatisfy:

UserAuthentication&Authorization

Thesystemshallallowstudents,instructors,andadministratorstologinsecurely.The system shall enforce role-based access control.

Usersshallonlyaccessfunctionalitiespermittedbytheirrole.

Exam Scheduling & Management

- Instructorsshallbeabletocreateexaminationsbydefiningcourse,title,duration,andexamtype.
- Thesystemshallsupportschedulingofexamswithspecificstartandendtimes.
- Exams shall be stored securely and managed centrally.

Systemmaintainsanexamscheduleforeachstudent(andinstructor).

Question Bank Management

- Instructorsshallbeabletoadd,edit,anddeletequestions.
- Thesystemshallsupportdifferentquestiontypes,suchasmultiple-choiceanddescriptive questions.
- Questionsshallbereusableacrossdifferentexams.

ExamDelivery

- Studentscanviewupcomingexams.
- Atscheduledtime,studentscanstarttheexamonline.
- Systemdisplaysexaminterface,handlesnavigationbetweenquestions,enforcestimelimit, autosaves periodically.

ExamSubmission&Grading

- Studentsshallbeabletosubmitexamsmanually.
- Thesystemshallautomaticallysubmitexamswhentimeexpires.
- Objectivequestionsshallbegradedautomatically.
- Subjectivequestionsshallbeavailableformanualgradingbyinstructors.

ResultPublication&Viewing

- Instructors/adminpublishexamresultsoncegradingcomplete.
- Studentscanviewtheirscoresandfeedback(ifany).

UserandCourseManagement

- Admincancreate,update, removeuseraccounts(students,instructors).
- Admin/instructorscanmanagecourselist,assigncoursestoinstructorsorstudents.

Audit / Logs (optional but recommended)

- Systemmaintainslogsofuseractions(examcreation,usermanagement,examattempts, submissions) for traceability.

3.2 Non-functionalRequirements

System-levelconstraintsandqualityattributes:

SecurityRequirements

- Thesystemshallusesecureauthenticationmechanisms.
- Userpasswordsshallbestoredinencryptedform.
- Accesstoexaminationdatashallberestrictedbyrole.
- Thesystemshalllogcriticalactivitiesforauditingpurposes.

PerformanceRequirements

- Thesystemshallsupportmultipleuserstakingexamssimultaneously.
- Responsetimeduringexamoperationsshallremainacceptable.
- Automaticsavingshallpreventdatalossduringconnectivityissues.

ReliabilityandAvailability

- ✓ Thesystemshallbeavailableduringscheduledexamperiods.
- ✓ Thesystemshallrecovergracefullyfromunexpected failures.
- ✓ Automaticsubmissionshallensureexamcompletionincaseoftimeout.

UsabilityRequirements

- ✓ Theuserinterfaceshallbesimpleandintuitive.
- ✓ Thesystemshallsupportcommonwebbrowsers.
- ✓ Instructionsandfeedbackshallbeclearlydisplayed.

MaintainabilityandScalability

- Thesystemshallbemodulartosupportfutureenhancements.
- Newfeaturesshallbeaddedwithoutdisruptingexistingfunctionality.

- Thesystemshallsupportfuturegrowthinusernumbers.

3.3 Constraints("Pseudo-requirements")

- Usersmusthavevalidcredentialsassignedbytheuniversitybeforeusingsystem(especially instructors/admin).
- Examschedulingmustbecoordinatedwithuniversityacademiccalendarandcommunicatedto students prior to exam.
- Foroffline/low-bandwidthusers:considerlimitations(e.g.autosaveintervals,minimalpage load data).
- Dataprivacy:students'gradesandpersonaldatamustremainconfidential;accessbased only on roles.
- Systemdeploymentenvironment:likelyuniversityservers(orcloud),withconstraintson storage, bandwidth, hosting policy.

3.4 SystemModels

3.4.1 Scenarios(ExampleUserScenarios/Use-caseNarratives)

Scenario1:StudentTakesanOnlineExamination

StudentStartsandSubmitsaScheduledOnlineExam

PrimaryActor:

Student

Stakeholders&Interests:

- Student:Wantstotaketheexamsmoothlyandsubmitanswerssuccessfully.
- University:Wantssecureandfairexaminationhandling.
- System:Mustenforceexamtime,authentication,andautomaticsubmission.

Preconditions:

1. Studentisregisteredinthesystem.
2. Studentisassignedtoacoursethathasascheduledexam.
3. Examdateandtimearealreadypublished.
4. Studenthassuccessfullyloggedintothesystem.

MainSuccessFlow(NormalFlow):

1. Studentlogsinusingusernameandpassword.

2. Systemverifiescredentialsandredirectsthestudenttothedashboard.
3. Studentnavigatesto“MyExams”.
4. Studentviewsthescheduledexam(e.g.,ComputerNetworksMidExam,10:00AM–11:00AM).
5. At10:00AM,the“StartExam”buttonbecomesactive.
6. Studentclicks“StartExam”.
7. Systemdisplaystheexaminterfacewith:
    - Timer(60minutescountdown)
    - Numberofquestions
    - Navigationpanel
    - Save/Next/Previousbuttons
8. Studentanswersallquestions(e.g.,40MCQs+3shortquestions).
9. Systemautosavesanswersevery30seconds.
10. Studentclicks“SubmitExam”beforethetimerends.
11. Systemconfirmssubmissionandlockstheexam.
12. SystemautomaticallygradesMCQquestionsandstorestheresults.
13. Studentreceivesconfirmation:“Yourexamhasbeensubmittedsuccessfully.”

AlternativeFlow(A1):InternetDisruptionDuringExam

1. Studentlosesconnectiontemporarily.
2. Systemactivatesofflineautosavemode(savinglocallyinbrowserstorage).
3. Whenconnectionisrestored,answerssyncautomaticallytotheserver.
4. Studentcontinuestheexamnormally.

Postconditions:

- Examsubmissionisstoredinthedatabase.
- MCQresultsareauto-graded.
- Essayquestionswaitforinstructorgrading.
- Studentcannotretakeormodifyanswers.

Scenario2:InstructorCreatesandPublishesanExam

InstructorCreates,Configures,andPublishesanExam

PrimaryActor:

Instructor

Stakeholders&Interests:

- Instructor:Wantstocreateexamcontentandmanageavailability.
- Students:Needaccuratescheduleandcorrectexamsettings.
- System:Mustenforceaccesscontrolanddataconsistency.

Preconditions:

1. Instructorhasavalidsystemaccount.
2. Instructorisassignedtoat leastonecourse.
3. Coursequestionbankisavailable(optionalbutrecommended).

MainSuccessFlow(NormalFlow):

1. Instructorlogsintothesystem.
2. Fromthedashboard,instructorselects“ExamManagement”.
3. Instructorclicks“CreateExam”.
4. Instructorentersexamdetails:
   - Course:DatabaseSystemsII
   - Exam Type: Mid Exam
   - Date/Time:Feb 08, 2026, 2:00PM– 3:00 PM
   - Duration:60minutes
   - Questionformat:40MCQs+5short answer
5. InstructorselectsquestionsfromthequestionbankORcreatesnewquestions.
6. Instructorreviewsexamsummary(questions,totalmarks,duration).
7. Instructorpresses“PublishExam”.
8. Systemvalidatesallfieldsandschedulestheexam.
9. Studentsenrolledinthecourseautomaticallyseetheexamin“My Exams”.
10. Instructorreceivesconfirmation:“Yourexamhasbeenpublishedsuccessfully.”

AlternativeFlow(A1):MissingRequiredFields

1. Instructorforgetstosettheexamduration.
2. Systemdisplayserrormessage:“Durationisrequired.”
3. Instructorentersthemissingduration.
4. Instructorresubmitstheform.

Postconditions:

- Examisstored,scheduled,andvisibletoallassignedstudents.
- Questionsarelocked(cannotbeeditedafterpublishingunlessrescheduled).
- Systemgeneratesnotificationstostudents.

3.4.2 UseCaseModel

Thefollowingisa**UseCaseDiagram**forthesystem,showingactorsandtheirinteractionswith the system. This gives a high-level view of all major functionalities as they relate to user roles.

Actors:

- Student
- Instructor(ExamCreator)
- Administrator

UseCases(majorones):

- Register/Login/Logout
- ViewExamSchedule
- TakeExam(StartExam)
- SubmitExam
- ViewResults
- CreateExam
- ManageQuestionBank(Add/Edit/DeleteQuestions)
- PublishResults
- ManageUsers
- Manageurses

Relationships/Notes:

- Everyactormustuse“Register/Login”beforeaccessingotherusecases.
- “TakeExam”isavailableonlyforStudent.
- “CreateExam”,“ManageQuestionBank”,“PublishResults”areforInstructor.
- “ManageUsers”,“ManageCourses”areforAdministrator.
- “ViewExamSchedule”and“ViewResults”arecommonforStudentandInstructor.

Actors&UseCases

Student ----------+

→Register/Login/Logout--<<include>>-------- +

→ViewExamSchedule

→TakeExam----------<<include>>-->LoadQuestions

→SubmitExam----------<<include>>-->AutoEvaluation

→ViewResults----------<<extend>> ------>DownloadResultSlip

Instructor(ExamCreator) ---------- +

→Register/Login/Logout--<<include>> ---------+

→ViewExamSchedule

→CreateExam

→ManageQuestionBank---------- >(Add/Edit/DeleteQuestions)

→ PublishResults

Administrator ----------+

→Register/Login/Logout--<<include>> ---------+

→ManageUsers

→ManageCourses

→ManageDepartments

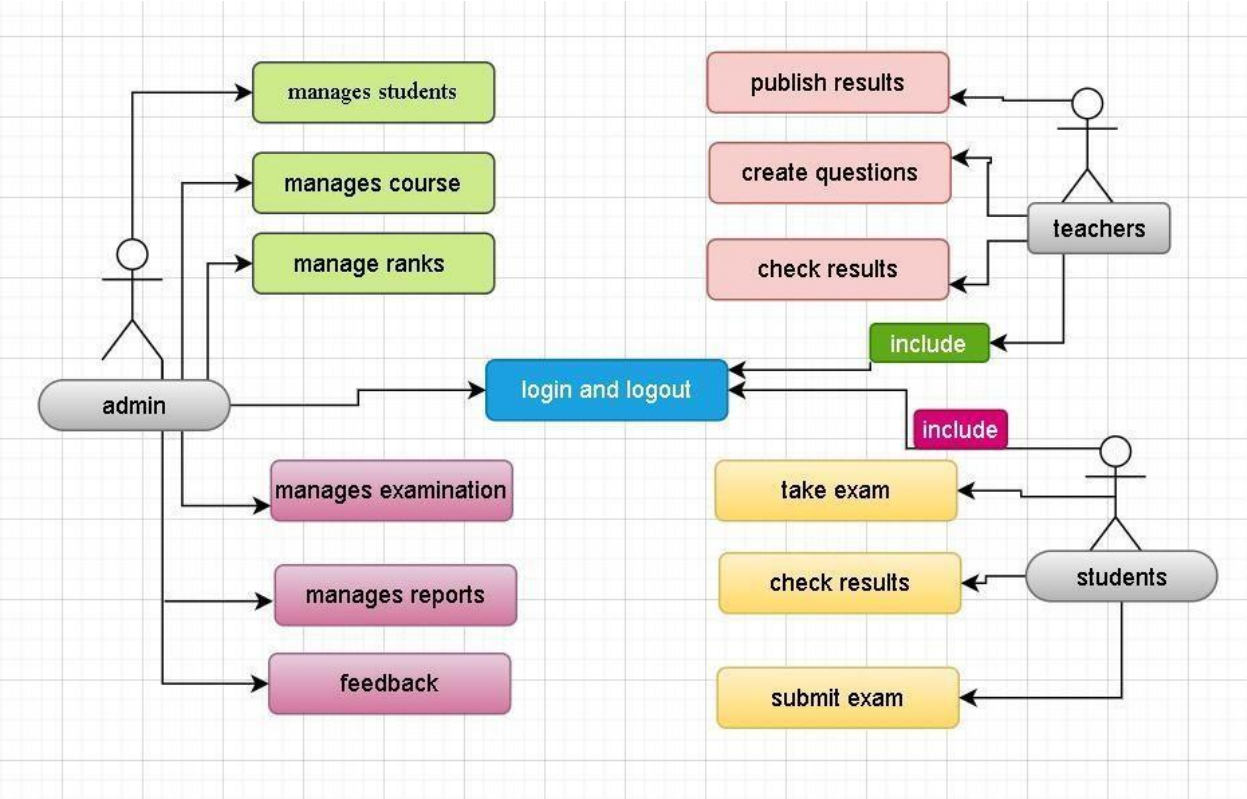
→BackupSystem

→ViewSystemLogs

System(Internal) ----------+

→AutoSubmitonTimeout------<<extend>>--------- TakeExam

→AutoEvaluation----------<<include>>--------SubmitExam



CHAPTERFOUR

SYSTEMDESIGNDOCUMENT(SDD)

4.1 Introduction

This chapter presents the **System DesignDocument (SDD)** for the Werab University Online Examination System (WR-OES). The SDD translates the requirements defined in previous chapters into a structured and detailed system design. It describes how the system is organized, how its components interact, and how design decisions support system functionality, security, scalability, and maintainability.

The SDD serves as a technical blueprint for developers and as a reference document for stakeholders, ensuring that the system design meets academic, functional, and institutional requirements.

4.2 PurposeoftheSystemDesignDocument

TheprimarypurposesoftheSystemDesignDocumentare:

- Todescribetheoverallstructureandarchitectureof WR-OES
- Todefinesystemcomponentsandtheirresponsibilities
- Todocumentdesigndecisionsandjustifychosenapproaches
- Toprovideaclearguideforsystemimplementationandtesting
- Toensureconsistencybetweenrequirementsandsystembehavior

Byformalizingthedesign,theSDDreducesambiguityandsupportssystematic development.

Themodelisderivedfromsystemrequirements,scenarios,andusecasesusing**Abbott'stechnique**, where:

- **Nouns**→classes
- **Verbs**→operations
- **Ownership**→aggregationorcomposition

Thisensuresconsistencybetween**requirements,usecases,andsystemdesign**.**Main**

Classes and Responsibilities

User(AbstractClass)

Representsallsystemusers.

Attributes

- userId
- name
- email
- password

Operations

- login()
- logout()

Subclasses

- Student
- Examiner
- Administrator

Inheritanceisusedtoavoidredundancyandsupportrole-basedbehavior.

Student

Representscandidateswhotakeonlineexaminations.

Responsibilities

- Registerforexams
- Takeexams
- Submitanswers
- Viewresults

Operations

- registerExam()
- startExam()
- submitAnswers()
- viewResult()

Astudentcanattemptmultipleexams.

Examiner

Representsacademicstaffresponsibleforcreatingandmanagingexams.

Responsibilities

- Createexams
- Addquestions
- Reviewresults

Operations

- createExam()
- addQuestion()
- publishExam()
- viewResults()

Anexaminercanmanagemultipleexams.

Administrator

Representssystemadministrators.

Responsibilities

- Manageusers
- Manageexams
- Performsystemmaintenance

Operations

- manageUsers()

- manageExams()
- backupSystem()

Exam

Representsanonlineexamination.

Attributes

- examId
- title
- duration
- totalMarks

Operations

- scheduleExam()
- closeExam()

Anexam**owns**questionsand submissions.

Question

Representsexamquestions.

Attributes

- questionId
- questionText
- marks
- questionType

Operations

- validateAnswer()

Questionscannotexistindependentlyofanexam(composition).

Option

Representsmultiple-choiceoptions.

Attributes

- optionId
- optionText
- isCorrect

Belongstoaquestion.

Submission

Representsastudent’sexamattempt.

Attributes

- submissionId
- startTime
- endTime

Operations

- submit()

Eachsubmissionbelongstoonestudentandoneexam.

Answer

Representsanswerssubmittedbystudents.

Attributes

- answerId
- selectedOption
- descriptiveAnswer

Belongs to a submission.

Result

Representsevaluationresults.

Attributes

- score
- grade

Operations

- generateResult()

Associatedwithasubmission.

Notification

Representssystem-generatedalerts.

Attributes

- message
- sentTime

Operations

- send()

RelationshipsBetweenClasses

- **Inheritance**:
  Student,Examiner,Administrator→User
- **Associations**:
  Student attempts Exam
  ExaminercreatesExam
- **Composition**:
  Exam → Question
  Question → Option
  Submission → Answer
- **System-Automated Classes**:
  Resultgenerationandnotificationsarehandledautomaticallybythesystem.

3.5 DataDictionary

Table:User

Field Name	DataType	Constraints	Description
user_id	INT	PK, AUTO_INCREMENT	UserID

name	VARCHAR(120)	NOTNULL	Fullname
email	VARCHAR(120)	UNIQUE	Email
password	VARCHAR(255)	NOTNULL	Hashedpassword
role	ENUM	NOTNULL	Student, Examiner, Admin

Table:Student

FieldName	DataType	Constraints
student_id	INT	PK,FK→User
student_number	VARCHAR(15)	UNIQUE
program	VARCHAR(100)	NOTNULL

Table:Examiner

FieldName	DataType	Constraints
examiner_id	INT	PK,FK→User
department	VARCHAR(100)	NOTNULL

Table:Exam

FieldName	DataType	Constraints
exam_id	INT	PK,AUTO_INCREMENT
title	VARCHAR(150)	NOTNULL
duration	INT	NOTNULL
total_marks	INT	NOTNULL
examiner_id	INT	FK→Examiner

Table:Question

FieldName	DataType	Constraints
question_id	INT	PK
exam_id	INT	FK→Exam
question_text	TEXT	NOTNULL

marks	INT	NOTNULL
type	ENUM	MCQ,Descriptive

Table:Option

FieldName	DataType	Constraints
option_id	INT	PK
question_id	INT	FK→Question
option_text	VARCHAR(255)	NOTNULL
is_correct	BOOLEAN	NOTNULL

Table:Submission

FieldName	DataType	Constraints
submission_id	INT	PK
student_id	INT	FK→Student
exam_id	INT	FK→Exam
start_time	DATETIME	NOTNULL
end_time	DATETIME	NOTNULL

Table:Result

FieldName	DataType	Constraints
result_id	INT	PK
submission_id	INT	FK→Submission
score	DECIMAL(5,2)	NOTNULL
grade	VARCHAR(5)	NOTNULL

3.5DynamicModel

WhatisaDynamicModel?

The **dynamic model** describes **how the system behaves over time**. Unlike the object model (which shows *what exists*), the dynamic model explains:

- **Howobjectsinteract**

- **Theorderofoperations**
- **Howsystemstateschange**
- **User-drivenandsystem-drivenbehavior**

Dynamicmodelingisessentialtounderstand**runtimebehavior**andtoverifythatsystemrequirements are correctly implemented.

Dynamicmodelsaretypicallyrepresentedusing:

- **SequenceDiagrams**
- **StateDiagrams**
- **ActivityDiagrams**

1. SequenceDiagrams–Explanation

Purpose

Sequencediagramsillustrate**interactionamongobjectsandinterfaces**ina**time-ordered manner**.

Theyanswer:

- Whointeractswithwhom?
- Inwhatorderdomessagesoccur?
- Howdoesthesystemrespondtouseractions?

1.1 StudentTakesanExam(SequenceDiagram)

Explanation

1. The**Student**initiatesinteractionbyloggingin.
2. The**LoginUI**validatescredentials.
3. Uponsuccess,the**DashboardUI**isdisplayed.
4. Thestudentselectsanexam.
5. The**ExamUI**loadsquestionsfromthesystem.
6. Thestudentsubmitsanswers.
7. Thesystemevaluatesanswersanddisplaysresults.

Significance

- Showsend-to-endstudentworkflow.
- Ensuresnostepisskipped(authentication→exam→result).
- Helpsdevelopersidentifyintegrationpoints.

1.2 ExaminerCreatesandPublishesExam

Explanation

1. The**Examiner**logsinandaccessesthedashboard.
2. Theexaminercreatesanewexam.

3. Questionsareaddedormodified.
4. Theexamispublishedandmadeavailabletostudents.

Significance

- Demonstratesexamlifecyclefromcreationtoavailability.
- Separatesexaminerresponsibilitiesfromstudentactions.
- Helpsenforceauthorizationrules.

1.3 SystemAuto-EvaluationProcess

Explanation

1. Theexamsystemreceivessubmittedanswers.
2. Theevaluationengineprocessesobjectivequestionsautomatically.
3. Scoresarecalculated.
4. Resultsarestoredandmade available.

Significance

- Showsinternalsystembehaviorwithoutuserinteraction.
- Highlightsautomation.
- Importantforperformanceandreliabilityanalysis.

2. StateDiagrams–Explanation

Purpose

Statediagramsrepresent**thelifecycleofanobjectorinterface**,showing:

- Possiblestates
- Transitionsbetweenstates
- Eventstriggeringtransitions

They answer:

- Whatstatescananobjectbein?
- Howdoesitmovefromonestateto another?

2.1 LoginInterfaceStateDiagram

Explanation

- Theinterfacestartsinthe**LoginScreen**state.
- Invalidcredentialsmovethesystemto**ErrorMessage**.
- Validcredentialstransitionthesystemtotheappropriatedashboard.

WhyImportant

- Preventsunauthorizedaccess.
- Handlesincorrectinputgracefully.
- Ensuressecurityflowiscorrect.

2.2 StudentDashboardStateDiagram

Explanation

- Thedashboardisidleafterlogin.
- Thestudentcaneither:
    - Viewexams
    - Viewresults
- Startinganexamexitsthedashboardstate.

WhyImportant

- Clearlyseparatesdashboardactivities.
- Preventsconflictingactions(e.g.,takingtwoexamsatonce).

2.3 ExaminerDashboardStateDiagram

Explanation

- Theexaminercancreateexams,addquestions,publishexams,orviewresults.
- Eachtaskreturnscontroltothedashboard.

WhyImportant

- Enforcescontrolledexammanagement.
- Supportsmodulartaskexecution.

2.4 AdministratorDashboardStateDiagram

Explanation

- Theadministratoraccessessystem-leveloperations.
- Usermanagement,exammanagement,andmaintenanceareisolatedtasks.
- Alltransitionsreturntothemaindashboard.

WhyImportant

- Maintainssystemstability.
- Ensuresadministrativeauthorityiscentralized.

2.5 ExamInterfaceStateDiagram

Explanation

- Theexamloadsquestions.
- Studentsnavigatebetweenquestions.
- Submissiontriggersconfirmationandevaluation.
- Resultsaregeneratedattheend.

WhyImportant

- Preventsprematuresubmission.
- Ensuresnavigationconsistency.
- Supportstime-boundexams.

2.6 ResultInterfaceStateDiagram

Explanation

- Resultsaredisplayedafterevaluation.
- Studentscanviewscoreandfeedback.
- Sessionendsafterlogout.

WhyImportant

- Ensurestransparency.
- Providesclosuretotheexamprocess.

3. ActivityDiagram–Explanation

Purpose

Activitydiagramsmodel**workflowandcontrolflow**.They

show:

- Decisionpoints
- Parallelactions
- Sequentialactivities

UIActivityFlowExplanation

1. Useropensthesystem.
2. Logincredentialsareentered.
3. Ifvalid→dashboardloads.
4. Role-basedactionsareperformed.
5. Resultsorconfirmationsaredisplayed.
6. Sessionends.

WhyImportant

- Givesahigh-leveloperationaloverview.
- Usefulforstakeholdersandnon-technicalreviewers.

OverallImportanceofDynamicModel

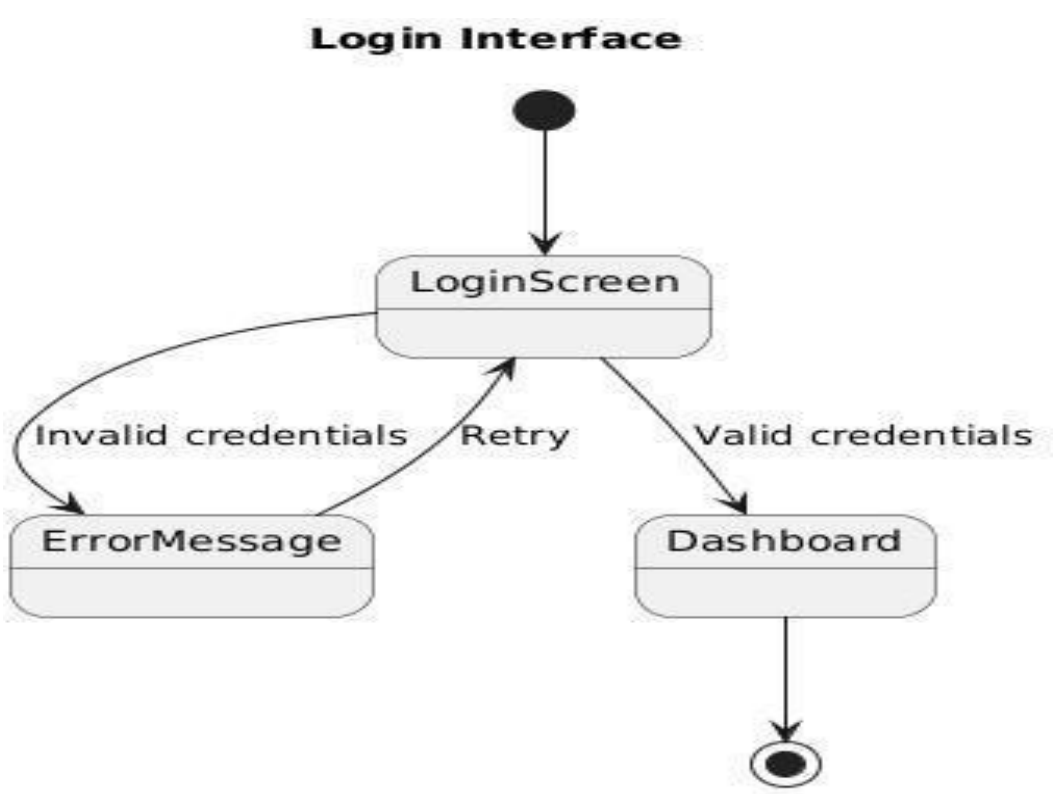
Aspect	**Contribution**
SequenceDiagrams	Interactionclarity
StateDiagrams	Lifecyclecontrol
ActivityDiagrams	Workflowunderstanding
UIFocus	Userexperiencemodeling

3.5UserInterface

TheOnlineExaminationSystemUIis**web-based**,role-oriented,anduser-friendly.

GeneralUIDesignPrinciples

- Simplenavigation
- Role-basedaccess
- Responsivedesign
- Securesessionhandling



1. LoginInterface

Purpose

TheLoginInterfaceisthe**entrypoint**totheOnlineExaminationSystem.Itensures**secureauthentication**andcontrolsaccessbased on user roles.

BehaviorExplanation

- Thesysteminitiallydisplaysthe**LoginScreen**.
- Usersentervalidcredentials(username/emailandpassword).
- Ifcredentialsare**invalid**,thesystemtransitionstoan**ErrorMessage**stateandpromptstheusertoretry.
- Ifcredentialsare**valid**,thesystemredirectstheusertotheappropriatedashboard(Student,Examiner,or Administrator).

Importance

- Preventsunauthorizedaccess.
- Supportsrole-basednavigation.
- Ensuressystemsecurityandaccountability.

2. StudentDashboard

Purpose

TheStudentDashboardprovidesstudentswithaccess to **examination-related activities**.

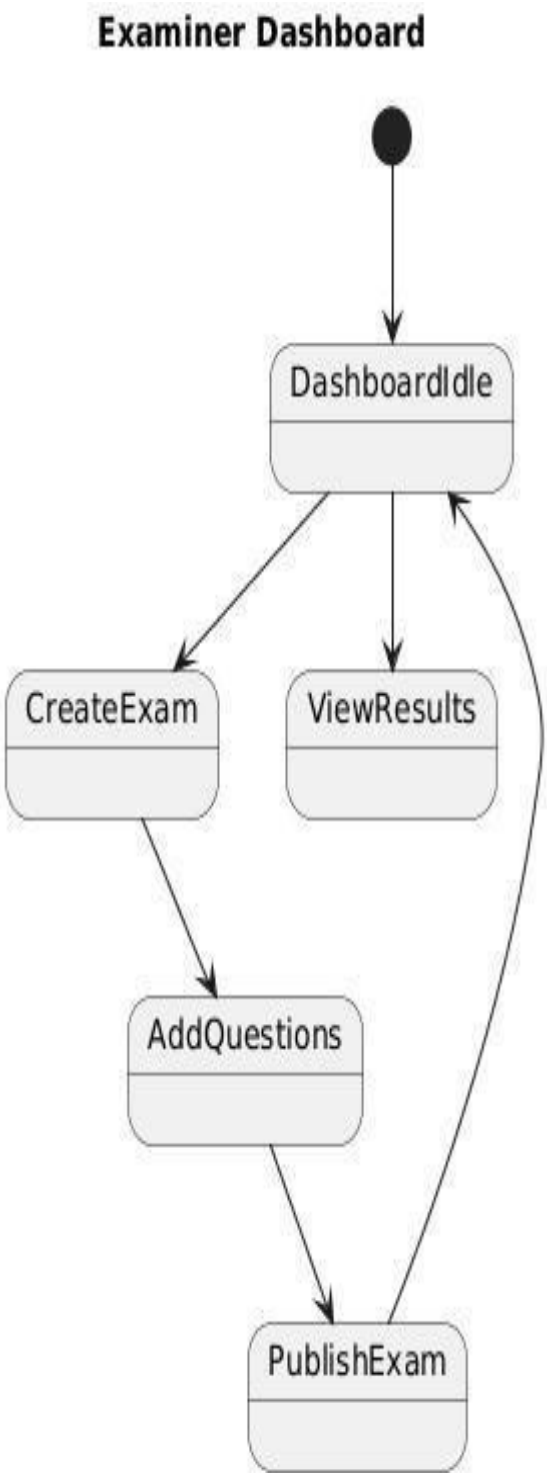


BehaviorExplanation

- Afterlogin,thestudententersthe**Dashboard Idle** state.
- Thestudentcan:
  - Viewavailableexams.
  - Startanexam.
  - Viewpreviouslyobtainedresults.
- Onceanexamisstarted,controlistransferredto the **Exam Interface**.

Importance

- Actsasacentralizedcontrolpanelfor students.
- Reducescomplexitybygroupingallstudent actions in one interface.
- Enhancesusabilityandlearningexperience.



3. ExaminerDashboard

Purpose

TheExaminerDashboardallowsinstructorsto **create,manage,andmonitorexaminations**.

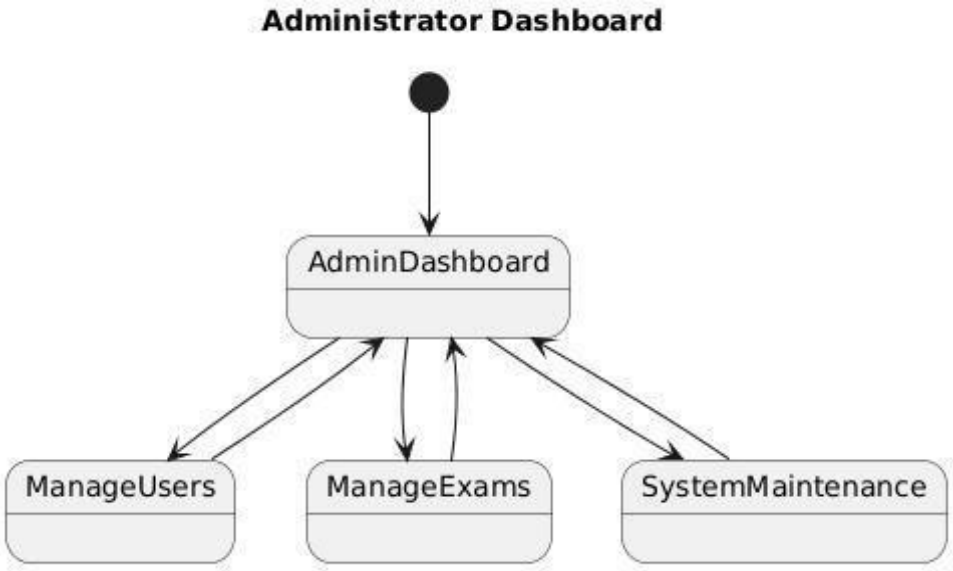
BehaviorExplanation

- Theexaminerentersthedashboardafter authentication.
- Theexaminercan:
  - Createnewexams.
  - Addormodifyquestions.
  - Publishexamsforstudents.
  - Viewexamresults.
- Aftercompletingtasks,thesystemreturnstothe dashboard state.

Importance

- Enablesefficientexammanagement.
- Ensurescontrolledpublicationofexams.
- Supportsassessmentmonitoring.
-

4. AdministratorDashboard



4. AdministratorDashboard

Purpose

TheAdministratorDashboardisresponsiblefor**system-widemanagementandmaintenance**.

BehaviorExplanation

- Theadministratorcan:
  - Manageusers(create,update,deactivate).
  - Manageexams.
  - Performsystemmaintenancetaskssuchasbackupsandmonitoring.
- Eachadministrativeoperationeventuallyreturnstothemaindashboard.

Importance

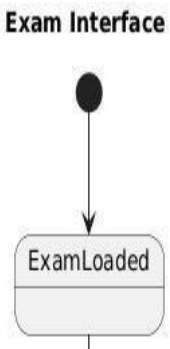
- Ensuressystemreliabilityandintegrity.
- Supportsscalabilityandmaintenance.
- Enforcessystempolicies.
-

5. ExamInterface

Purpose

TheExamInterfaceisthe**coreoperational interface** where students take exams.

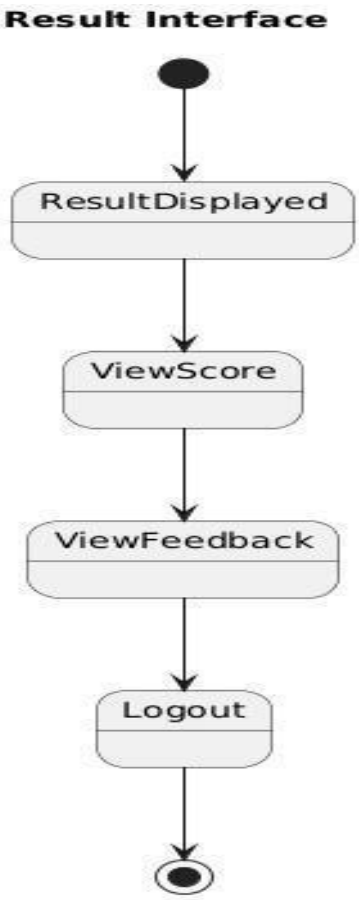
BehaviorExplanation



- Whenanexamstarts,thesystemloadsthe exam data.
- Questionsaredisplayedsequentially.
- Studentscan:
    - Navigatebetweenquestions.
    - Reviewpreviousanswers.
    - Submittheexamatanytimebefore expiration.
- Uponsubmission,thesystemconfirms submission and generates results.

Importance

- Ensuressmoothexamdelivery.
- Preventsdatalossthroughcontrolled transitions.
- Supportsfairandtime-boundassessment.



Accessibility&Usability

- Keyboardnavigation
- Clearinstructions
- Lowbandwidthsupport

6. ResultInterface

Purpose

TheResultInterfacepresents**examoutcomes**tostudents.

BehaviorExplanation

- Afterevaluation,resultsaredisplayed.
- Studentscan:
  - Viewtheirscore.
  - Readfeedback(ifavailable).
  - Logoutorreturntothedashboard.
- Theinterfaceensuresclarityandtransparency.

Importance

- Providesimmediatefeedback.
- Enhanceslearningthroughperformancereview.
- Maintainsacademictransparency.

Chapter5

SystemDesignDocument(SDD):

1. Introduction

The **Online Examination System (OES)** is a web-based platform designed to automate the entire examination process—from exam creation and scheduling to delivery, grading, and reporting.Traditionalpaper-basedexamsfacechallengessuchasmanualerrors,logistical overhead,delayedresults,andlimitedscalability.TheOESaddressestheseissuesbyprovidinga secure, efficient, and transparent digital solution.

Thesystemsupportsmultipleuserroles:

- **Students**:Takeexamsonline,receiveinstantfeedback,andtrackperformance.
- **Instructors**:Createexams,managequestionbanks,monitorexamsessions,andevaluate results.
- **Administrators**:Overseeusermanagement,enforceexampolicies,andgenerateinstitutional reports.

Thisdocumentoutlinesthe**architecture,designgoals,andsystemcomponents**toensurescalability, maintainability, and security.

1.1 PurposeoftheSystem

ThepurposeoftheOESisto:

- **Centralizeexammanagement**:Provideasingleplatformforexamcreation,scheduling, and evaluation.
- **Enhanceaccessibility**:Allowstudentstotakeexamsfromanywhereusingsecurelogin credentials.
- **Improveefficiency**:Automategradingforobjectivequestionsandstreamlineevaluation for subjective ones.
- **Ensurefairnessandtransparency**:Randomizequestions,enforcetimelimits,andprevent cheating through monitoring mechanisms.
- **Supportinstitutionaldigitaltransformation**:AlignHaramayaUniversity'sacademic processeswithmoderntechnology.

1.2 DesignGoals–OnlineExaminationSystem(OES)

UsabilityandAccessibility

- Thesystemmustprovideanintuitive,user-friendlyinterfaceforallroles(students,instructors, administrators).
- Accessibilityfeaturesincluderesponsivedesignacrossdevices(desktop,tablet,mobile), multilingual support, and compliance with accessibility standards (e.g., WCAG).

- Clearnavigation,examtimers,andreal-timefeedbackensureminimalconfusionduring high-stress exam sessions.

Scalability

- Thearchitecturemustsupportthousandsofconcurrentexamtakerswithoutperformance degradation.
- Horizontalscaling(addingservers)andverticalscaling(upgradingresources)shouldbe possible.
- Loadbalancingandcachingmechanismsensuresmoothoperationduringpeakexamperiods.

ModularityandMaintainability

- Thesystemisdecomposedintoindependentmodules:
  - **UserManagement**:registration,authentication,roleassignment.
  - **ExamCreation**:questionbank,scheduling,randomization.
  - **ExamDelivery**:secureexaminterface,timer,proctoring.
  - **Grading&Reporting**:automatedscoring,analytics,feedback.
- Modulardesignsimplifiesdebugging,testing,andfutureenhancements.

PerformanceandReliability

- Criticaloperations(login,examsubmission,grading)mustbeoptimizedforspeedand reliability.
- Thesystemshouldguaranteeuptimeduringscheduledexams,supportedbyredundancy and failover mechanisms.
- Real-timemonitoringensuresquickdetectionandresolutionofissues.

SecurityandDataProtection

- Strongauthentication(multi-factorlogin,sessionmanagement)preventsunauthorizedaccess.
- Encryption(HTTPS,databaseencryption)protectssensitiveexamdata.
- Anti-cheatingmechanisms:IPtracking,randomizedquestions,sessionmonitoring,and optional AI-based proctoring.
- Auditlogsensureaccountabilityandtraceabilityofallactions.

ConsistencyandDataIntegrity

- Examresults,logs,andsubmissionsmustbestoredaccuratelyandconsistently.
- Transactionaldatabaseoperationspreventdatalossduringfailures.

- Automatedbackupsandrecoverymechanismssafeguardagainstsystemcrashes.

SupportforIncrementalDevelopment

- ThesystemdesignsupportsAgileandincrementaldevelopment.
- Newfeatures(e.g.,adaptivetesting,AIproctoring,advancedanalytics)canbeintegrated without disrupting existing modules.

Definitions,Acronyms,andAbbreviationsforan**OnlineExamination System(OES)System Design Document**. Terminology, capitalization, and descriptions are standardized to align with **IEEE SDD conventions**, without changing meaning or disrupting existing modules.

1.3 Definitions,Acronyms,andAbbreviations–OnlineExaminationSystem

Term / Acronym	Description
OES	**OnlineExaminationSystem**–Aweb-basedplatformthatautomatesexamination creation, delivery, grading, and result reporting.
SDD	**SystemDesignDocument**–Aformaldocumentdescribingthesystemarchitecture, components, interfaces, and design decisions.
MCQ	**MultipleChoiceQuestion**–Aquestiontypeconsistingofpredefinedanswer options from which the student selects one or more correct answers.
ExamSession	**ScheduledExaminationPeriod**–Adefinedtimewindowduringwhichauthorized students can access and complete an examination.
Proctoring	**MonitoringMechanism**–Supervisionofexaminationactivitiestoprevent academicdishonesty;mayinvolveautomated(AI-based)orhumanoversight.
Instructor	**SystemUserRole**–Responsibleforcreatingexaminations,managingquestion banks, and reviewing or evaluating student results.
Student	**SystemUserRole**–Aregistereduserwhotakesexaminationsonline,submits responses, and views grades or performance reports.
Administrator	**SystemUserRole**–Managesuser accounts,configuresexaminationpolicies, andoverseesoverallsystemoperations.
Frontend	**Client-SideInterface**–Thepresentationlayerbuiltusingtechnologiessuchas HTML, CSS, and JavaScript; handles user interaction and display.
Backend	**Server-SideLogic**–Theapplicationlayerimplementedusingtechnologiessuchas PHP or Node.js; processes business logic and examination workflows.
Database	**StructuredDataRepository**–Storesexaminationdataincludingquestions, responses,results,userprofiles,andauditlogs(e.g.,MySQLorPostgreSQL).

3-Tier Architecture

AgileScrum

Software Design Pattern – An architectural model that separates the system into Presentation, Application, and Data layers.

Iterative Development Methodology – A project management framework using short development cycles (sprints) to incrementally deliver system features.

Authentication	**IdentityVerification**–Theprocessofvalidatingusercredentialsbeforegranting system access.
Authorization	**AccessControlMechanism**–Determineswhichsystemactionsandresources an authenticated user is permitted to access.
Encryption	**DataProtectionTechnique**–Securesexaminationdataanduserinformation during storage and transmission.
QuestionBank	**CentralizedRepository**–Acollectionofexaminationquestionsorganizedby subject, difficulty level, and question type.
Randomization	**ExaminationSecurityFeature**–Shufflesquestionsand/oransweroptionsto minimizecheating.
Grading Engine	**AutomatedEvaluationModule**–Instantlyscoresobjectivequestionsandsupports manual grading for subjective responses.
Analytics	**PerformanceAnalysisModule**–Providesstatisticalinsightsonstudent performance, exam difficulty, and institutional outcomes.

Chapter6

3. ProposedSoftwareArchitecture

3.1 Overview

The OES is designed following **system design principles** outlined in Chapter 6. It decomposes the system into manageable subsystems, each with clear responsibilities and interfaces. The architecture emphasizes **modularity, reliability, and user-friendliness**, ensuring exams can be created, delivered, and evaluated securely.

3.2 SubsystemDecomposition

Subsystemsaredefined bytheservicestheyprovide:

- **AuthenticationSubsystem**→login,sessionvalidation,roleassignment.
- **ExamManagementSubsystem**→scheduling,questionbank,randomization.
- **ExamDeliverySubsystem**→timedinterface,navigation,auto-save.
- **AnswerProcessingSubsystem**→capture,validation,submission.
- **GradingSubsystem**→automaticscoring,manualevaluation,resultstorage.
- **ReportingSubsystem**→analytics,charts,downloadablereports.
- **SecuritySubsystem**→accesscontrol,activitylogging,integrityenforcement.

3.3 Hardware/SoftwareMapping

- **ClientNode**:JavaFXdesktopapplicationrunningonstudent/instructormachines.
- **DatabaseNode**:MySQLserverforpersistentstorage.
- **JavaRuntime+JavaFXSDK**:ExecutionenvironmentforGUIandlogic.
- **NetworkConnectivity**:Securecommunicationbetweenclientanddatabase.

3.4 PersistentDataManagement

- **UserData**:credentials,roles,profiles.
- **ExamData**:questions,schedules,configurations.
- **AnswerData**:auto-savedresponses,finalsubmissions.
- **ResultData**:scores,feedback,analytics.
- **Logs**:audittrailsforintegrityandrecovery.

3.5 AccessControlandSecurity

- **Role-basedaccess**(student,instructor,admin).
- **Sessionmanagement**topreventconcurrentlogins.
- **Encryption**ofsensitivedata.
- **Randomization**ofquestions/optionstoreducecheating.
- **Auditlogging**foraccountability.

3.6 GlobalSoftwareControl

- **Startupsequence**:loadlogin,initializedashboard.

- **Examflowcontrol**:managetimer,navigation,submission.
- **Errorhandling**:user-friendlymessages,exceptionlogging.
- **Statemanagement**:trackcurrentquestion,timeremaining,answerstatus.

3.7 BoundaryConditions

- **Initialization**:systemloadsconfigurationandavailableexams.
- **Termination**:securelogout,savestate,closeconnections.
- **Failurehandling**:auto-submissionontimeout,recoveryfromcrashusingauto-save.
- **Shutdown**:ensurealllogsandresultsarepersisted.

4. SubsystemServices–OnlineExaminationSystem

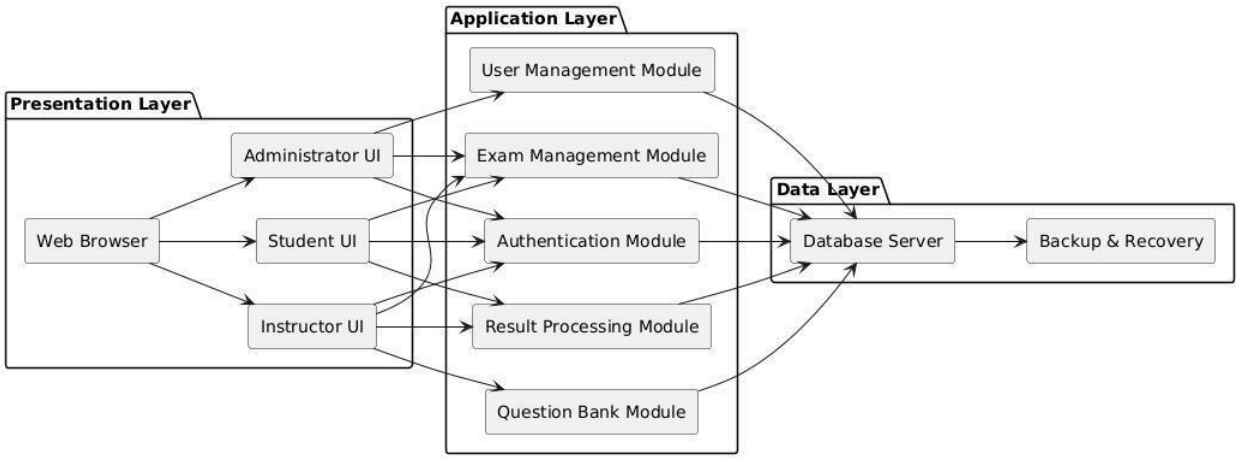
Eachsubsystemexposeswell-definedservices(interfaces)toothersubsystems:

- **AuthenticationSubsystem**
  - login(username,password) validateSession()
  - getUserRole()
- **ExamManagementSubsystem**
  - createExam() scheduleExam(date,duration)
  - loadQuestions(examID)
- **ExamDeliverySubsystem**
  - startExam(examID) displayQuestion(index)
  - autoSaveAnswer(questionID,answer)
- **AnswerProcessingSubsystem**
  - submitExam() validateAnswers()
  - storeResponses()
- **GradingSubsystem**
  - gradeMCQ() evaluateEssay()
  - generateScoreReport()
- **ReportingSubsystem**
  - displayResults(studentID) generateAnalytics(courseID)
  - exportReport(format)

- **SecuritySubsystem**
    - randomizeQuestions() restrictNavigation()
    - logActivity(event)

Theproposedarchitecture forthe Werabe**UniversityOnlineExaminationSystem(WR-OES)** followsa**three-tierlayeredarchitecture**to ensurescalability,security,and maintainability.

- **PresentationLayer:**Providesuserinterfacesforstudents,instructors,andadministrators through aweb browser.It handles user interactions such as login, exam access, and result viewing.
- **Application Layer:** Contains the core business logic of the system, including authentication, exam management, question bank handling, result processing, and user management. This layer processes requests from the presentation layer and enforces system rules.
- **Data Layer:** Manages persistent data storage using a database server and ensures data safety through backup and recovery mechanisms. All application modules interact with this layer for storing and retrieving information.



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